

DSA 210 TERM PROJECT

MACHINE LEARNING METHODS

Machine Learning Models

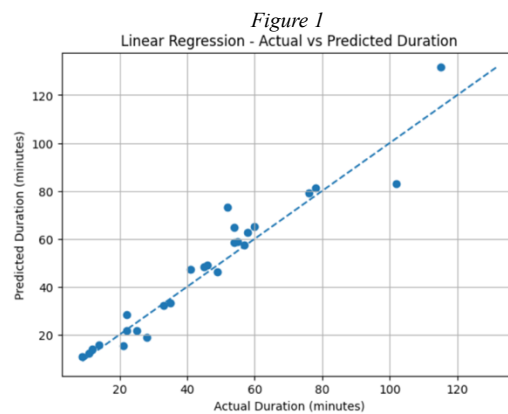
To predict actual travel duration, several machine learning models were implemented and compared.

1. Linear Regression

This model achieved the best performance with an R^2 score of 0.914, explaining approximately 91.4% of the variance in travel duration. The average prediction error was around 5.46 minutes, indicating strong predictive capability.

MAE: 5.46
RMSE: 7.76
 R^2 : 0.914

Meaning of R^2 : This model explains approximately 91.4% of the variation in actual travel duration.
Comment: This indicates high predictive power.



2. Random Forest

This model also performed well with an R^2 score of 0.909 and a similar error level (~5.36 minutes), showing that non-linear relationships exist but do not significantly outperform linear modeling.

MAE: 5.36
RMSE: 7.97
 R^2 : 0.909

Meaning of R^2 : This model explains approximately 90.9% of the variation in actual travel duration.
Comment: This indicates high predictive power.

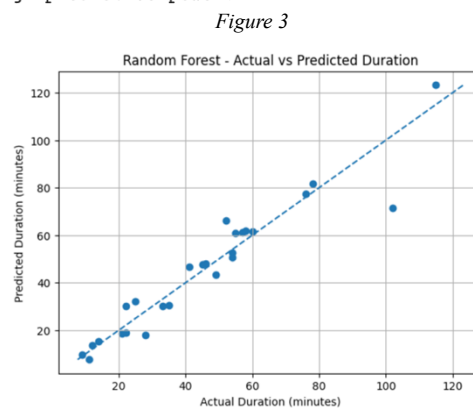


Figure 4

3. K-Nearest Neighbours (kNN)

KNN achieved an R^2 score of 0.90, performing slightly worse than the top models but still demonstrating strong predictive power.

MAE: 5.83

RMSE: 8.34

R^2 : 0.9

Meaning of R^2 : This model explains approximately 90.0% of the variation in actual travel duration.

Comment: This indicates high predictive power.

Figure 5

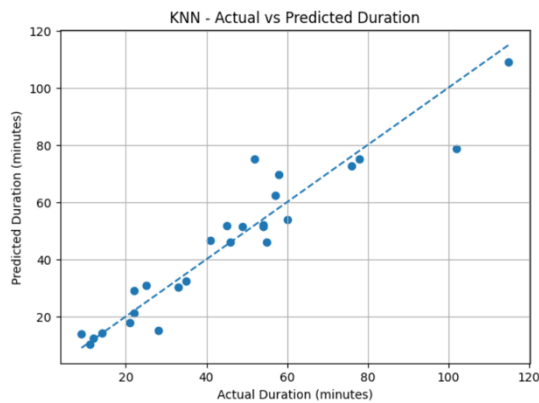


Figure 6

4. Decision Tree

This model had the lowest performance ($R^2 = 0.784$), suggesting that a single tree structure is not sufficient to capture the complexity of the data.

MAE: 8.0

RMSE: 12.27

R^2 : 0.784

Meaning of R^2 : This model explains approximately 78.4% of the variation in actual travel duration.

Comment: This indicates high predictive power.

Figure 7

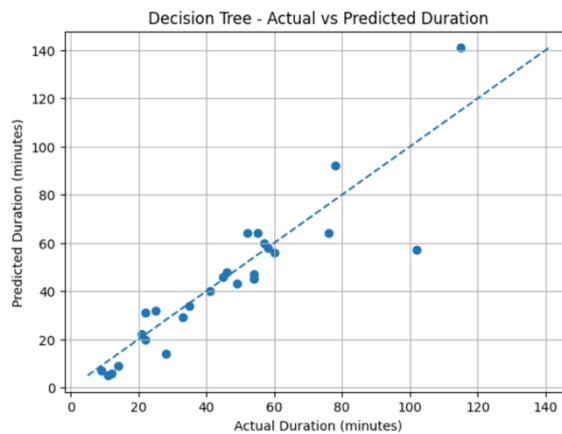


Figure 8

Overall, Linear Regression and Random Forest emerged as the most effective models, indicating that travel duration can be predicted with high accuracy using relatively simple features such as distance, traffic level, and estimated duration.